NAUSICAA CSA: design, readout and response to ion tracks

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Abstract

This note describes the working design of the NAUSICAA charge-sensitive amplifier and the calculation used to reconstruct ions arriving in successive readout frames. A slow feedback discharge, nominally 20 s, and voltage averages over 5 ms give a reconstructed frame noise of 51.26–58.95 elementary charges RMS. Complete pad histories are used to calculate accumulated voltage, compression and noise. Pilot samples compare 2, 3 and 4 mm pad pitches with 20, 30 and 40 ms frames. The reconstructed ion matrices are saved for the track analysis.

1 The working circuit

Each metal pad is connected to a charge-sensitive amplifier (CSA). The ion current changes the charge on the feedback capacitor C_f and therefore the output voltage. Negative feedback keeps the input voltage nearly constant. With sufficient amplifier gain, an input charge Q changes the output by approximately $-Q/C_f$. The transistor calculation determines the actual response, including finite gain, bandwidth and feedback discharge.

Figure **??** shows the circuit. The pad drives the gate of the PMOS input transistor M_1. A change in its gate voltage changes its current. The M_2 and M_3 branch establishes the input-stage operating point; M_4 carries the signal into the output branch, whose PMOS load is formed by M_5 and M_6. The output is connected back to the input through C_f and M_f. The capacitance integrates charge. The small current through M_f restores the baseline continuously.

The circuit uses 180 nm CMOS models, a supply of 1.8 V, $C_f = 3$ fF and an external input capacitance $C_{\text{ext}} = 1.25$ pF. This external value is a design input held fixed throughout the comparison. The simulated temperature is 293.15 K. At the quiescent operating point the output is 0.886 V, the supply current is 210.7 µA and the corresponding circuit power is 0.379 mW per channel.

Table 1: Transistor dimensions from the simulated v3 netlist. Multiplicity counts parallel devices with the listed dimensions.

Device	Type	W (μm)	L (μm)	Multiplicity
M_1	PMOS	220	0.5	1
M_2	NMOS	27	0.5	3
M_3	NMOS	200	2	1
M_4	NMOS	2.88	1.44	1
M_5	PMOS	220	0.5	1
M_6	PMOS	1	6	1
M_f	NMOS	0.36	1200	1

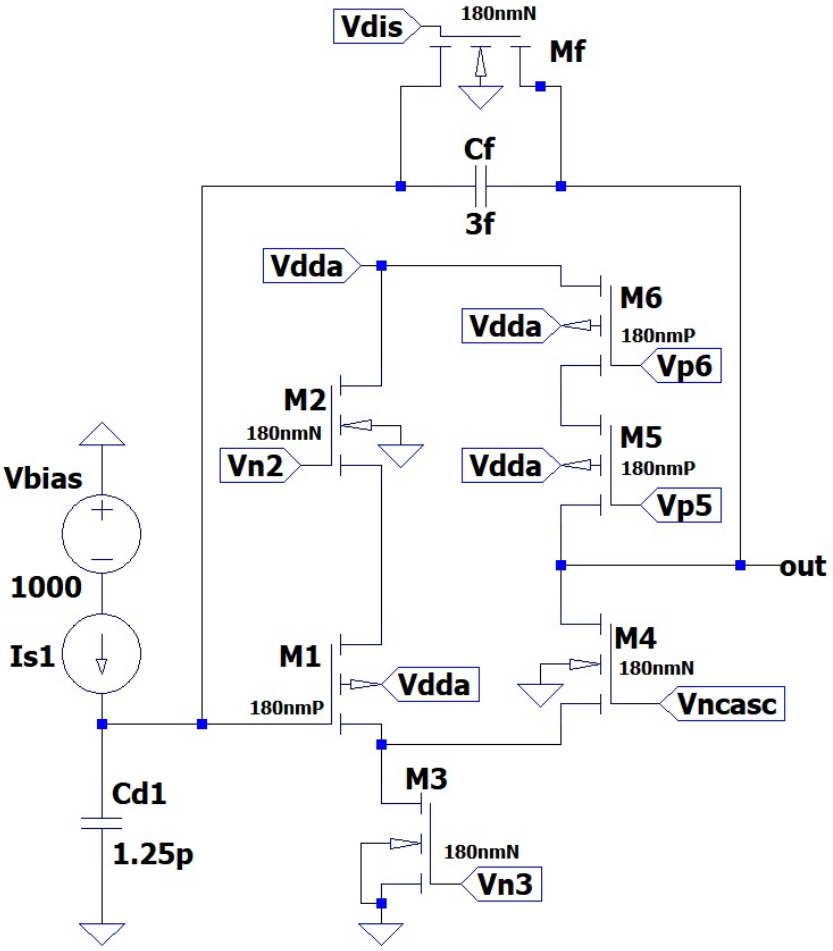


Figure 1: CSA topology supplied by Rogelio Palomo. The input source represents the pad current. The external input capacitance is C_{ext}; the transistor models add their own capacitances. Device dimensions and operating voltages used in the calculation are given in Tables ?? and ??.

Table 2: Bias voltages, rounded for display. The saved netlists retain the values used in the simulation.

Node	Voltage (V)
`Vdda`	1.8
`Vnmos`	1.6
`Vnload`	0.5385
`Vp6`	1.1
`Vp5`	0.8
`Vncasc`	0.6621
`Vdis`	0

2 Why the feedback discharge is slow

The input current from one ion is short. A pad can collect many ions over a much longer interval, and its CSA integrates their currents continuously. Between arrivals the feedback transistor removes some of the accumulated charge. Its conductance therefore affects both signal retention and noise.

The selected feedback device has a long, narrow channel. Near the operating point its current is very small, producing a slow return to baseline. The small-signal tail is fitted by

$$V(t) - V_0 = B \exp(-t/\tau_f). \tag{1}$$

The fit gives $\tau_f = 19.54\,\mathrm{s}$; the working design is labelled "20 s retention". This is a fitted property of the transistor circuit. Its response at large accumulated voltage is calculated directly in SPICE.

Figure **??** shows the retention scan. Increasing the feedback length reduces its contribution to the filtered noise. The improvement becomes smaller beyond the selected point. A longer discharge time also retains more charge from earlier buckets, increasing the accumulated voltage. The full pad histories are needed to check the resulting compression.

A retention time of 20 s does not set the duration of a readout frame. Voltage readings continue throughout the event, at intervals of 20, 30 or 40 ms. The signal from earlier charge is accounted for when differences between readings are converted into bucket counts.

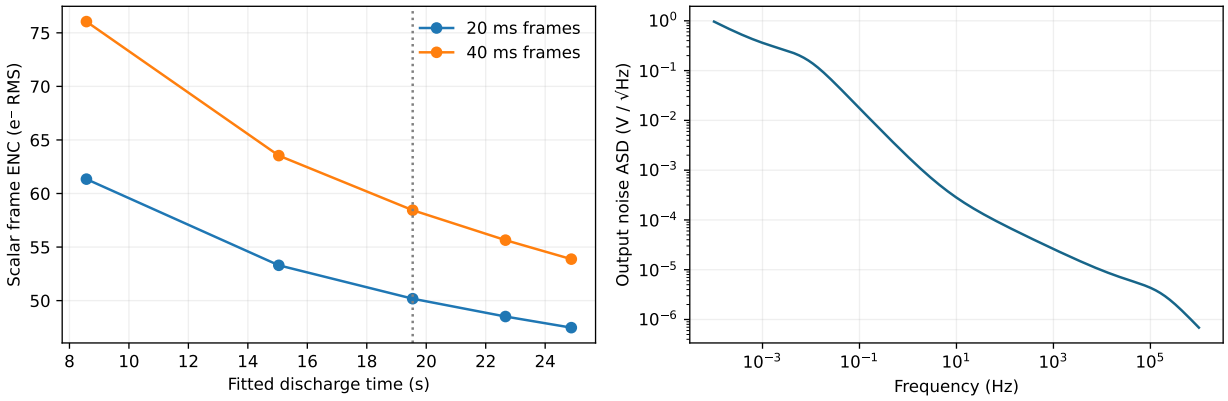


Figure 2: Left: the saved retention scan, with 5 ms averaging and a feedback-corrected frame difference. These ENC values precede the three-frame reconstruction described below. Each point uses its fitted discharge time and its own operating-point calibration. The vertical line marks the selected circuit. Right: its one-sided output noise amplitude spectral density from SPICE.

3 From bucket counts to the pad current

The detector simulation supplies $N_{p,n}$, the number of positive ions collected by pixel p in bucket n. Each xenon molecular ion, Xe_2^+, carries charge $+e$. A bucket is an interval of collection times. For this calculation the arrivals are distributed uniformly within it.

The current from one ion follows from the weighting potential ϕ_w [**?**]. This potential is calculated without free charge, with the selected pad at unit potential and the other conductors at zero. For the benchmark used here, the pad is a square of side 0.5 mm in an infinite grounded plane. The gas is homogeneous above the plane and the potential tends to zero at infinity. On the central axis, for pad half-width a,

$$W(z) = -\frac{\mathrm{d}\phi_w}{\mathrm{d}z} = \frac{4a^2}{\pi(z^2 + a^2)\sqrt{z^2 + 2a^2}}. \tag{2}$$

This expression follows from the solid-angle potential of a square electrode; its derivation is recorded in the single-ion appendix of `notes/physics/nausica.tex`, using Ref. [**?**].

Take collection at $t = 0$. The ion approaches the centre at a constant speed $v = 3\,\mathrm{m/s}$ through the final gap $g = 5\,\mathrm{mm}$. With positive current delivered into the input,

$$i_1(t) = evW(-vt), \qquad -g/v \le t < 0, \qquad \int i_1(t)\,\mathrm{d}t = Q_g = e[1 - \phi_w(g)]. \tag{3}$$

The finite-path integral is retained. Collection ends the current. The actual IFG and surface dielectrics require a more detailed weighting-field calculation; the present geometry fixes the input-current benchmark for all three pilots.

For frame boundaries t_n, width $T = t_n - t_{n-1}$, and uniform mean arrival times, the input to the circuit is

$$i_p(t) = \sum_n \frac{N_{p,n}}{T} \int_{t_{n-1}}^{t_n} i_1(t - u)\,\mathrm{d}u. \tag{4}$$

The integration variable u is the collection time. Thus induction can begin before the bucket in which the ion reaches the pad. Consecutive buckets drive one continuous transient, including the zero-ion buckets between deposits.

4 Voltage readings and charge reconstruction

4.1 Averaging and frame differences

Let $V(t)$ be the CSA output relative to its quiescent baseline. At each frame boundary t_n, the readout uses the voltage averaged over a centred window of width $T_a = 5\,\mathrm{ms}$:

$$\overline{V}_n = \frac{1}{T_a} \int_{t_n - T_a/2}^{t_n + T_a/2} V(t)\,\mathrm{d}t. \tag{5}$$

A physical readout can form this mean from repeated voltage samples over the window. The present calculation performs the average on the SPICE transient. Figure **??** shows its timing.

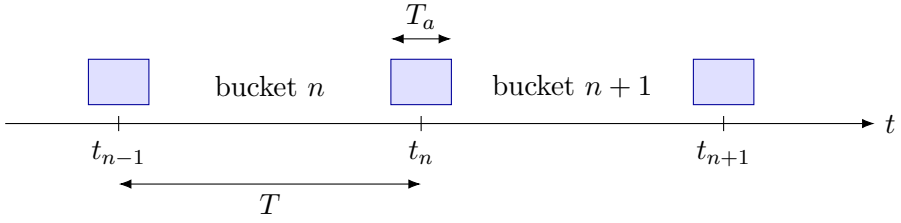


Figure 3: Voltage-averaging windows at successive frame boundaries. An average combines samples within each shaded window. The difference uses two adjacent averages. The drawing is schematic.

Earlier charge leaves a voltage that decays between boundaries. Define $\alpha = \exp(-T/\tau_f)$. The measured frame difference is

$$d_n = -(\overline{V}_n - \alpha \overline{V}_{n-1}), \qquad y_n = \frac{d_n}{g}. \tag{6}$$

Here g, in volts per ion, is obtained from the SPICE response to a uniform packet in one frame: it is that packet's central-frame d_n divided by its ion count. The sign makes positive ion charge give positive y_n.

For an ideal, rapidly responding CSA and negligible decay, the output is a step for an instantaneous charge input. Averaging that step over T_a gives a ramp. The difference between two such ramps

gives a trapezoidal charge weight: a rising edge, a flat interval and a falling edge. Thus averaging and frame differencing provide the shaping used in this calculation. The finite circuit response and α determine its calibrated form.

4.2 Removing contributions from neighbouring frames

A centred averaging window overlaps both sides of a frame boundary. Induction also starts before collection, and the amplifier takes time to respond. Consequently a packet contributes to neighbouring y_n values. Calibrating only its central value leaves a positive bias when all frames are summed.

The saved small-signal packet response supplies three coefficients,

$$y_n = b_{\text{late}} N_{n-1} + N_n + b_{\text{early}} N_{n+1}. \tag{7}$$

The central coefficient is one by the definition of g. The coefficient b_{late} describes the response left by charge collected in the preceding frame; b_{early} describes the response before charge is collected in the following frame. Their numerical values and the longer tail are checked against the saved circuit response.

Write Eq. (**??**) as $\mathbf{y} = M\mathbf{N}$, with a tridiagonal M. The reconstructed counts are

$$\widehat{\mathbf{N}} = M^{-1}\mathbf{y}. \tag{8}$$

The same solve is applied to the mean signal and the sampled noise. It removes the linear frame mixing while retaining compression in the transistor response. The saved interval includes 2 quiet frames before the event and 3 s of recovery afterwards. These quiet frames provide the boundary conditions. Signed estimates are kept, including negative noise fluctuations.

5 Noise in the reconstructed counts

The transistor models supply thermal and flicker noise. Flicker noise becomes larger towards low frequency, with a power spectrum approximately proportional to $1/f$. SPICE propagates these device sources through the circuit and gives the total output spectrum at the quiescent operating point. Let $S_V(f)$ be its one-sided power spectral density, in V^2/Hz.

The average and frame difference in Eqs. (**??**) and (**??**) give the squared voltage weighting

$$|F(f)|^2 = \underbrace{\left[\frac{\sin(\pi f T_a)}{\pi f T_a}\right]^2}_{\text{average}} \underbrace{\left[(1-\alpha)^2 + 4\alpha \sin^2(\pi f T)\right]}_{\text{frame difference}}. \tag{9}$$

High-frequency fluctuations are reduced by averaging. Slowly varying offsets largely cancel in the difference. Both operations must be included when noise is compared with the measured ion signal.

The covariance of the scalar ion estimates, for integer frame lag ℓ, is

$$C_y(\ell) = \frac{1}{g^2} \int_{f_{\min}}^{f_{\max}} S_V(f) |F(f)|^2 \cos(2\pi f \ell T)\, \mathrm{d}f, \tag{10}$$

with $f_{\min} = 0.0001\,\mathrm{Hz}$ and $f_{\max} = 1\,\mathrm{MHz}$. The covariance after reconstruction is

$$C_{\widehat{N}} = M^{-1} C_y M^{-T}, \qquad \mathrm{ENC}_n = \sqrt{(C_{\widehat{N}})_{nn}}. \tag{11}$$

The counts are expressed in elementary-charge units, so this standard deviation is the equivalent noise charge (ENC), in electrons RMS. Equation (**??**) and its numerical quadrature are implemented

Table 3: Calibration from the selected circuit. g converts the scalar frame difference into ion counts. ENC_y precedes the inverse and $\mathrm{ENC}_{\widehat{N}}$ follows it. r_1 is the interior lag-one correlation after reconstruction. ENC values are in elementary charges RMS.

Pitch (mm)	Frame (ms)	τ_f (s)	g (μV/ion)	ENC_y	$\mathrm{ENC}_{\widehat{N}}$	r_1
2	20	19.54	40.16	50.18	51.26	-0.329
3	30	19.54	41.04	54.63	55.32	-0.297
4	40	19.54	41.48	58.44	58.95	-0.274

in `bucket_noise_covariance.jl`; the inverse and covariance transformation are implemented in `ion_csa_correction.jl`.

The quoted ENC is the interior-frame value. The exact covariance for each saved history includes its boundary frames. Gaussian noise is sampled with this covariance, independently for each pad. A stable per-pad seed makes the realisation reproducible. The stationary operating-point noise spectrum is used for every mean history, including histories with compression.

6 Pilot comparison

Each configuration uses the same 100 deterministically spaced event IDs from each sample of 8000 double-beta and 8000 single-electron events. The three configurations share the sampled primary ionisation and event placement. Changing pitch also changes the bucket duration: 2, 3 and 4 mm correspond to 20, 30 and 40 ms. Their effects are therefore compared together.

Every pad first receives a calibrated linear response. Direct SPICE transients replace it where large accumulated voltage can cause compression, and for control histories spanning the predicted voltage range. The directly simulated range expands until the remaining control error is below 0.1 ENC. Independent solver refinement checks the selected transients. This procedure concentrates the transistor calculation on the histories that need it.

6.1 Signal relative to noise

For each positive input pixel–frame cell,

$$(S/N)_{p,n} = \frac{\widehat{N}_{p,n}^{\,\mathrm{noiseless}}}{\mathrm{ENC}_{p,n}}. \tag{12}$$

All such cells enter the distributions, including weak deposits. A cell-weighted mean gives equal weight to each cell. An event-weighted mean first averages within each event, then gives equal weight to each event.

Table 4: Signals and computational coverage. "Pads" counts complete pad histories over the selected events; SPICE is the number calculated directly. Pitch is in mm. Ions/cell uses positive input cells.

Pitch	Sample	Events	Ions/cell	Cell S/N	Event S/N	Pads	SPICE
2	Double beta	100	64.9	1.27	1.53	30026	889
2	Single electron	100	55.2	1.08	1.36	36378	644
3	Double beta	100	164.0	2.96	3.46	15012	1536
3	Single electron	100	138.6	2.50	3.03	18222	1291
4	Double beta	100	308.8	5.21	5.91	9307	1754
4	Single electron	100	258.5	4.36	5.09	11423	1634

The 2 mm configuration has been set aside for the present design because of its lower S/N. The remaining choice is between 3 and 4 mm. The coarser grid collects more charge in each cell and

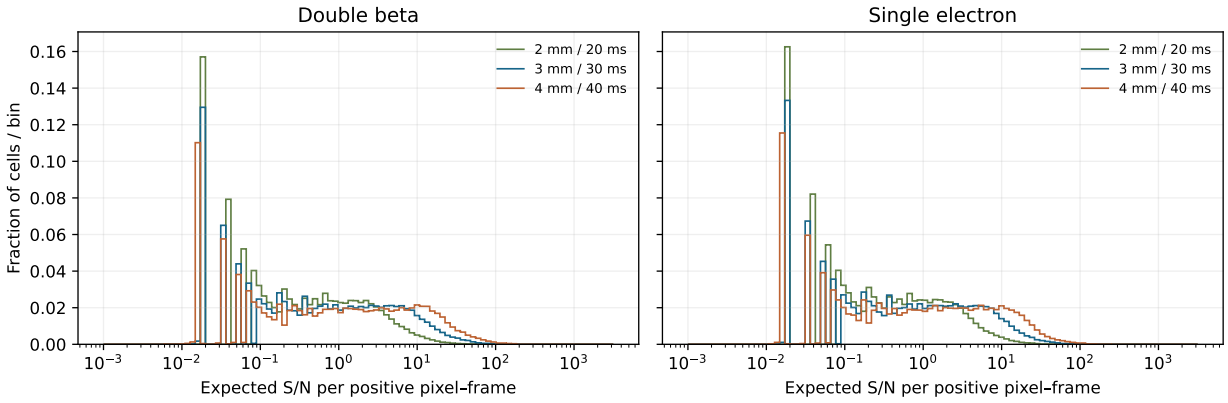


Figure 4: Expected S/N distributions from the saved corrected pilots. Every positive input pixel–frame cell contributes. The plotted bins use a logarithmic S/N axis; nonpositive values remain in the saved results.

improves S/N, while the finer grid samples the track more closely. Track continuity and the endpoint blobs will determine which gives the better reconstruction. S/N is a quality measure; the pilot applies no S/N acceptance cut.

6.2 Compression and total charge

Compression is measured by comparing the noiseless CSA estimate with the linear prediction for the same input. A local-loss flag requires the chosen fractional loss and a deficit of at least one frame ENC. A pad history is counted once when any positive-input bucket meets the criterion. An affected event contains at least one such pad history.

The compression deficit and total-charge bias describe different quantities. For event input $N_{\mathrm{tot}} = \sum_{p,n} N_{p,n}$, they are

$$D = \frac{\sum_{N_{p,n}>0} \max(\widehat{N}^{\mathrm{linear}}_{p,n} - \widehat{N}^{\mathrm{CSA}}_{p,n}, 0)}{N_{\mathrm{tot}}}, \tag{13}$$

$$B = \frac{\sum_{p,n} \widehat{N}^{\mathrm{CSA}}_{p,n} - N_{\mathrm{tot}}}{N_{\mathrm{tot}}}. \tag{14}$$

The signed sum in B includes quiet and recovery frames. Table **??** also gives its mean and event-to-event standard deviation after sampled noise is added.

Table 5: Mean event charge diagnostics. Local-loss events contain at least one positive cell with $\geq 50\%$ response loss and a deficit of at least one ENC. The bias uses the signed total over all saved buckets. These flags identify local distortion; the events are retained. Pitch is in mm.

Pitch	Sample	Local loss events (%)	Deficit (%)	Bias (%)	Noisy bias (%)	Spread (%)
2	Double beta	0.0	0.055	-0.089	+0.297	6.08
2	Single electron	0.0	0.043	-0.070	-0.719	5.57
3	Double beta	0.0	0.192	-0.290	+0.373	4.78
3	Single electron	0.0	0.164	-0.237	+0.116	4.27
4	Double beta	6.0	0.589	-0.677	-0.023	3.03
4	Single electron	7.0	0.512	-0.581	-0.747	3.88

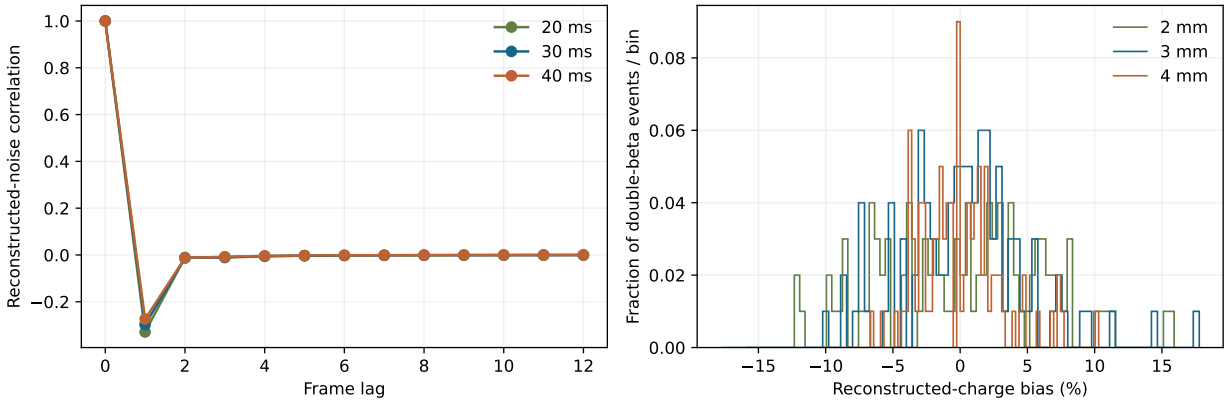


Figure 5: Left: temporal correlation of the reconstructed frame noise. Right: signed total-charge bias after noise is added, for the selected double-beta events.

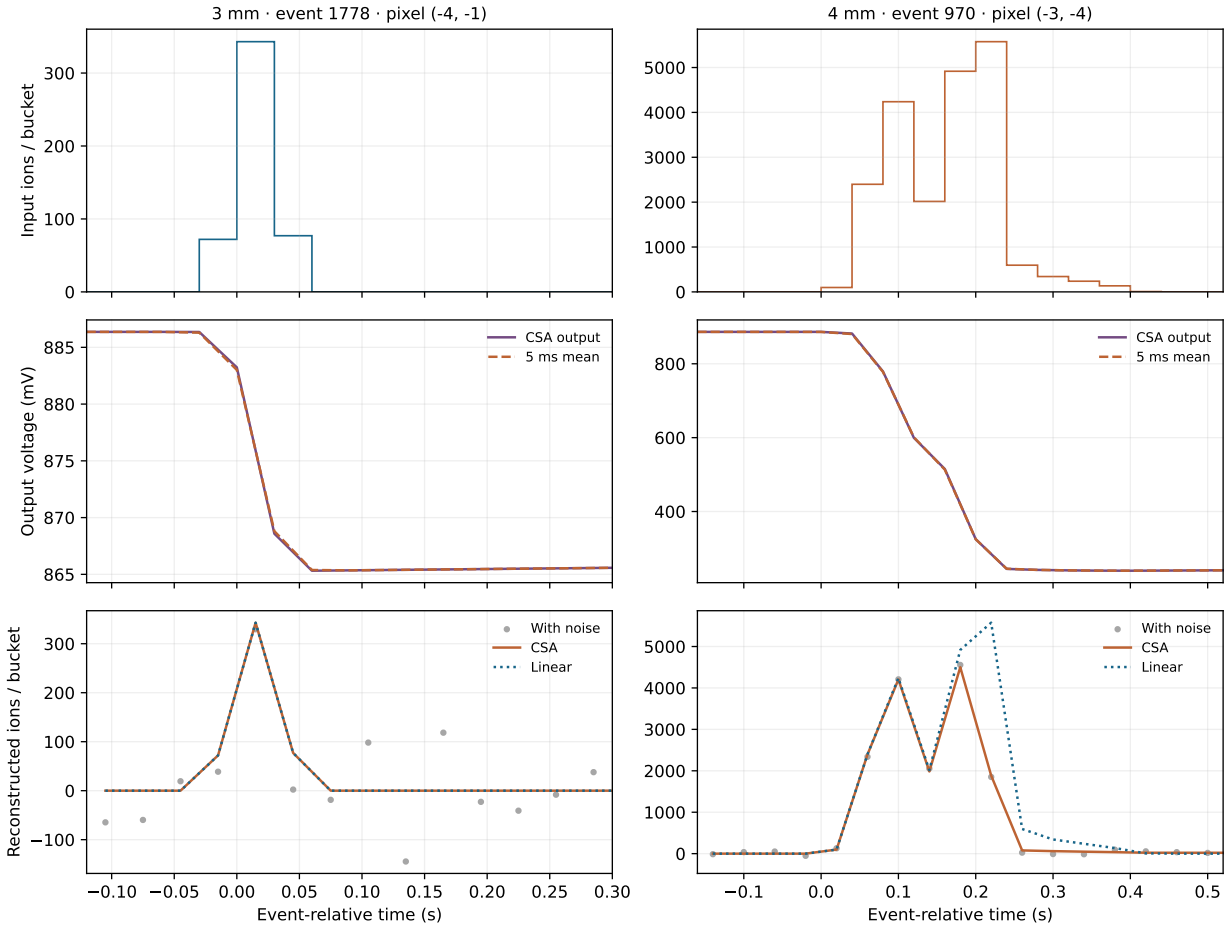


Figure 6: Examples from the saved double-beta pilots: a typical 3 mm history (left) and a compressed 4 mm history (right). The rows show input counts, CSA voltage and reconstructed counts. Empty buckets and the accumulated voltage are included. The figures show the event interval; the files also retain the recovery window.

7 Saved reconstruction and numerical controls

Each event is saved as a compact list of pixels with input charge, with a complete time vector for every listed pixel. The matrix has one row per pixel and one column per bucket in HDF5. Explicit pixel coordinates locate its rows; they need not form a rectangular spatial grid. An isolated deposit far from the main track adds its own pixels without adding the intervening pixels.

The output preserves input counts, the linear reference, the noiseless CSA estimate, the noise realisation, the final reconstructed counts and the voltage averages. Zero-input buckets within these histories and signed reconstructed values are retained. Absent pixels received no input charge; their electronic noise was not simulated. Measured zero suppression can be studied later using the saved signals and an explicitly defined readout area.

Every response records the original event IDs, signed pixel and bucket indices, geometry, event placement, input-file hash, circuit and reconstruction cards, calibration, implementation identity and noise seeds. HDF5 files are written atomically and marked complete after the requested events and ion ledgers pass validation.

Table 6: Numerical response controls after reconstruction. Errors are measured relative to the corresponding frame ENC.

Pitch (mm)	Maximum fast-response error (ENC)	Solver refinement (ENC)
2	0.026	3.82e-06
3	0.0308	3.92e-06
4	0.0269	1.09e-05

The note reduction checks 38 matched-input and summary conditions, including identical event IDs and primary-ion counts across geometries. Additional checks cover the input-current integral, spectral quadrature, covariance convergence and positive definiteness, noise sampling, frame alignment, exact input-ion accounting, and Float32 output precision against Float64 calculations. Each event begins at the quiescent operating point; charge accumulated in preceding unrelated events is a separate operating condition.

A Covariance and noise sampling

Equation (**??**) gives a Toeplitz covariance before the inverse, with entries $(C_y)_{ij} = C_y(|i-j|)$. A Cholesky factor L gives a noise realisation $\boldsymbol{\eta} = L\mathbf{z}$, with independent standard-normal components in $\mathbf{z}$. The stored reconstructed noise is $M^{-1}\boldsymbol{\eta}$. Its covariance is Eq. (**??**). The finite history and the quiet boundary frames make the final frame variances slightly position dependent; the analysis uses those variances.

The covariance integration uses the saved one-sided spectrum. High-frequency truncation for off-diagonal terms is accepted only when the omitted variance bounds the covariance error within the configured tolerance. The diagonal retains the full noise-band integral. Independent quadrature and zero-input noise samples validate the resulting matrices.

B Reproducing the note

The calculation follows the repository workflow: Julia performs the numerical reduction and writes HDF5; Python produces the tables, macros and figures. The note card is `config/specs/runs/csa_design_note.toml`. It selects the completed corrected pilot indexes and the saved retention scan. From the repository root, using the project's Julia scientific depot:

```
julia --project=. scripts/production/reduce_csa_design.jl
work/event-viewer-venv/bin/python \
  scripts/production/present_csa_design.py
```

The numerical summary and source hashes are saved in `results/csa_design/`. Generated tables, values, figures and a standalone copy of this source are in `latex_generated/csa_design/`. Presentation can be regenerated from the saved summary alone.

Each pilot has a run card and a reconstruction card. For example, the 2 mm calculation is reproduced with

```
work/event-viewer-venv/bin/python \
  scripts/production/run_ion_csa_pilot.py \
  --config config/specs/runs/ion_csa_v3_pilot_2mm.toml
work/event-viewer-venv/bin/python \
  scripts/production/run_ion_csa_correction.py \
  --config config/specs/runs/ion_csa_v3_corrected_pilot_2mm.toml
```

The corresponding 3 and 4 mm cards select their saved inputs. The production entry point gathers the six corrected pilot responses under the common run card and writes their manifests:

```
julia --project=. scripts/production/run_ion_csa.jl \
  config/specs/runs/ion_csa_v3_pilot_all.toml
```

The production schema is `nausicaa.ion_csa_response.v1`. `read_event` in `src/ion_signal/ion_csa_pipelin` returns an owned `ReconstructedIonEvent`, with input, mean, noise and reconstructed matrices indexed by (frame, pad) in Julia. The physical HDF5 axes are (pad, frame); the covariance reader supplies the matching frame covariance. The output metadata records the input-based channel selection and the treatment of absent pixels.

The companion HTML displays provide individual histories, severity selectors, distributions and numerical controls. Large HDF5 and simulation waveforms remain outside Git; the scripts, cards, source manifests and report artifacts are tracked.